



INTERNAL SECURITY

MODULE - 16

CYBER SECURITY

Cyber security can be defined as the protection of systems, networks and data in cyber space. It refers to the preventive methods used to protect information from being stolen, compromised or attacked.

4 Increasing inter connectivity and accessibility to computer-based systems that are critical to country's economy are further adding to the risk.

3 Security of critical infrastructure is essential for economic security



1 Difficult due to the diffused and varied nature of the threats.



2 Absence of tangible perpetrators

CYBER SECURITY

CYBER THREATS

CYBER CRIME

- Use of cyber space, i.e. computer, internet, cell phone, other technical devices, etc., to commit a crime by an individual or organised group is called cyber crime.
- Cyber attackers exploit the weakness in the software and hardware design through the use of malware
- **E.g.:** DoS attacks, hacking, identity theft.

CYBER WARFARE

- While there is no agreed definition of cyber warfare but when any state initiates the use of internet based invisible force as an instrument of state policy to sabotage and espionage against another nation, it is called cyber war.
- Although there is no clinching evidence of the involvement of a state in these attacks, it is widely held that in these attacks, non-state actors may have been used by state actors.

CYBER SECURITY

CYBER CRIME

1. CRIMES THAT TARGET COMPUTERS DIRECTLY



- Spreading computer viruses
- Denial-of-Services (DoS) attack is an attempt to make a machine or network resource unavailable to its intended users. It temporarily or indefinitely interrupts or suspends services of a host connected to the internet.
- Malware (malicious code) is a software used to disrupt computer operation, gather sensitive information, or gain access to private computer systems.
- It can appear in the form of code, scripts, active content, and other software.
- Malware is a general term used to refer to a variety of forms of hostile or intrusive software, for example Trojan Horses, rootkits, worms, adware etc.

CYBER SECURITY

CYBER CRIME

2. CRIMES FACILITATED BY COMPUTER NETWORKS OR DEVICES, THE PRIMARY TARGET OF WHICH IS INDEPENDENT OF THE COMPUTER NETWORK OR DEVICE

- **Economic fraud** to destabilise the economy of a country, attack on banking transaction system, extract money through fraud, acquisition of credit/debit card data, financial theft and intellectual theft of property
 - **Impairing the operations of a website** or service through data alteration, data destruction.
 - **Spreading pornography**
 - **Copyright infringement**
- **Cyber stalking**, outraging modesty of women, obscene content to humiliate girls and harm their reputation
 - **Threatening e-mail** assuming fake identity, virtual impersonation
 - **Breach of right to privacy**
 - **Misuse of social media** in fanning intolerance, instigating communal tensions and inciting riots
 - **Information warfare**
 - **Phishing scams**

CYBER SECURITY

KEY TERMS OF CYBER ATTACK

PHISHING

- Phishing is the act of attempting to acquire information, such as **usernames, passwords and credit card details** (and sometimes, indirectly, money) by **masquerading as a trustworthy entity** in an electronic communication.
- Communications purporting to be from popular social websites, auction sites, banks, online payment processors or IT administrators are commonly used to lure unsuspecting public.
- Phishing emails may contain links to websites that are infected with malware.
- Phishing is typically carried out by **email spoofing or instant messaging**.

VISHING (VOICE PHISHING)

- The term is a combination of 'voice' and 'phishing'.
- When phishing is done with the help of **telephonic system**, it is called vishing.



CYBER SECURITY

KEY TERMS OF CYBER ATTACK



TABNABBING

- It is one of the latest phishing technologies.
- It **takes advantage of tabbed browsing** (which uses multiple open tabs) that a user uses and silently redirects the user to the affected site.
- This technique operates in reverse to most phishing techniques as **it does not directly take the user to the fraudulent site**, but, instead, phishers load their fake page in one of the open tabs.



WHALING

Several recent phishing attacks have been directed specifically at **senior executives and other high profile targets** within businesses, and the term whaling has been coined for these kinds of attacks.



CYBER SECURITY

KEY TERMS OF CYBER ATTACK

RANSOMWARE

- Where all the information/data of the computer is hacked and the hacker demands certain amount of money in exchange for access to your own data.
- If ransom is not paid, then the data is deleted forever.
- E.g.: Petya, Wannacry



SPOOFING

- It is a situation in which one person or program successfully masquerades as another by falsifying data and thereby gaining an illegitimate advantage.
- The purpose of this is usually to fool programmes, systems or users into revealing confidential information, such as user names and passwords, to the attacker.



CYBER SECURITY

KEY TERMS OF CYBER ATTACK

BOTNETS

- A "bot," short for "robot," is a type of software application or script that performs tasks on command, allowing an attacker to take complete control remotely of an affected computer.
- A botnet is a collection of internet connected programs communicating with other similar programs in order to perform tasks.

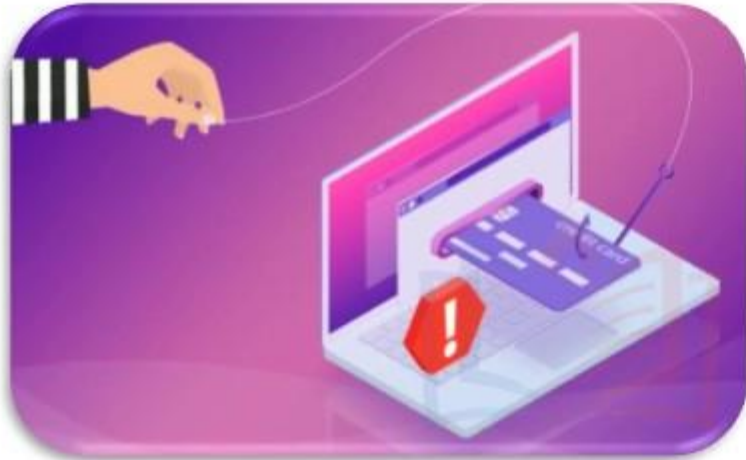


ZOMBIES

- The compromised machine may also be referred to as a "zombie."
- A zombie is a computer connected to the internet that has been compromised by a hacker, computer virus or trojan horse.
- It can be used to perform malicious tasks under remote direction.
- Botnets of zombie computers are often used to spread email spam and launch denial-of-service attacks.
- Most owners of zombie computers are unaware that their system is being used in this way. Because the owner tends to be unaware, these computers are metaphorically compared to zombies.

CYBER SECURITY

KEY TERMS OF CYBER ATTACK



DRIVE-BY

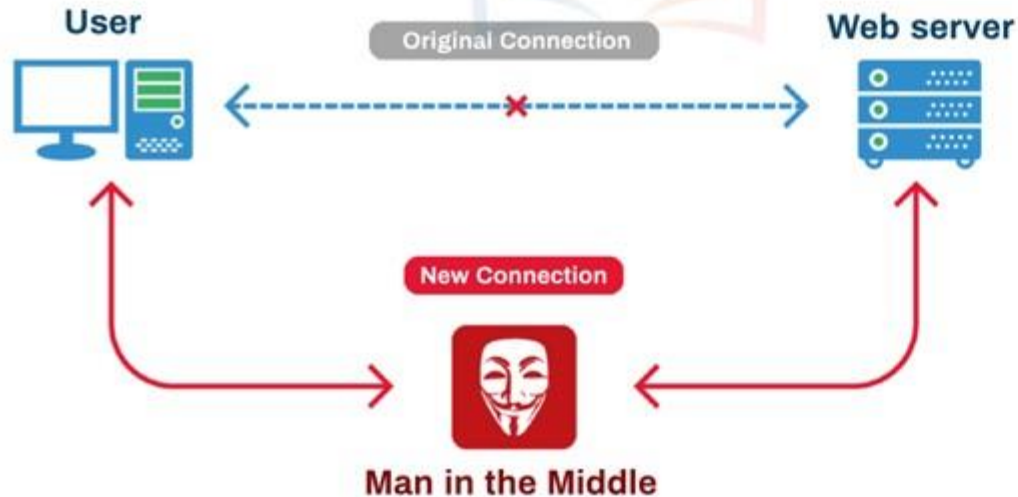
- Drive-by downloads are a common method of spreading malware.
- Cybercriminals look for insecure web sites and **plant a malicious script into HTTP** or PHP code on one of the pages.
- This script may install malware directly onto the computer of someone who visit the site.
- Such attacks are called 'drive-by downloads' because **they require no action on the part of the victim** beyond simply visiting the compromised web site.
- They are infected automatically if their computer is vulnerable in some way (e.g. if they have failed to apply a security update to one of their applications).

CYBER SECURITY

KEY TERMS OF CYBER ATTACK

MITM

Man in the middle attack is an attack where a middleman impersonates each endpoint and is thus able to manipulate both victims.



SPAM

- The **unsolicited sending of bulk email for commercial purposes**, is unlawful in some jurisdictions.
- While anti-spam laws are relatively new, limits on unsolicited electronic communications have existed for some time.



CYBER SECURITY

SPECIAL FEATURES OF CYBER WAR COMPARED TO TRADITIONAL WAR

1 INDEPENDENT THEATRE OF WAR

- The development of the internet and low-cost wireless communication is the contemporary equivalent of what airplanes were a hundred years ago.
 - Their use in economic, social and political transaction has increased at a rate that far exceeds the growth in airplane use over the last century.
 - There are signs that they have been used for aggressive purposes by some states.
 - There is also ample evidence of their use by criminals and terrorist groups.
 - It is only a matter of time, like air power a hundred years ago, before cyber space becomes an independent theatre of war.
- There is one important nuance in the treatment of cyber space as a fifth potential theatre of war, along with land, sea, air and space.
 - The use of cyber space depends on physical facilities like undersea cables, microwave and optical fibre networks, telecom exchanges, routers, data servers, and so on.
 - Protecting or attacking these is in the domain of the traditional arms of the military.
 - Cyber space is about attacks that compromise the capability to use these facilities.

CYBER SECURITY

SPECIAL FEATURES of CYBER WAR COMPARED TO TRADITIONAL WAR

2

AN UNDEFINED SPACE (NO SPECIFIC AREAS)



This means that a country's Government must ensure coherence between its security policy and the diplomatic stance taken by it in multilateral and bilateral discussions on matters like internet and telecom governance, human rights related to information freedoms, trade negotiations on infotech services, and so on.

- The defence of cyber space has a special feature.
- The national territory or space that is being defended by the land, sea and air forces is well defined.
- Outer space and cyber space are different.
- They are inherently international even from the perspective of national interest.
- Hence, national defence and international cooperation are inevitably intermeshed.

CYBER SECURITY

SPECIAL FEATURES of CYBER WAR COMPARED TO TRADITIONAL WAR

3 DISGUISED ATTACKERS

- There is another feature of cyber space that complicates the design of security structures and policies compared to the other theatres of conflict.
- In cyber space, it is very easy for an attacker to cover his tracks and even mislead the target into believing that the attack has come from somewhere else.
- This difficulty in identifying the perpetrator makes it difficult to rely on the capacity to retaliate as a deterrent.

4 NO CONTACT WAR

The evolution of technology impacts the nature of conflict and war.



CYBER SECURITY

CRYPTOCURRENCY AND THEIR LINKAGES TO TERROR FINANCING

WHAT IS A CRYPTOCURRENCY?



It is decentralised digital currency without a central bank or single administrator that can be sent from user-to-user on the peer-to-peer network without the need for intermediaries.



CYBER SECURITY

CRYPTOCURRENCY AND THEIR LINKAGES TO TERROR FINANCING

HOW TO GET THEM?

- Exchange with local currency
- Mine
- Sell something and get them in return.



ARGUMENTS AGAINST THEM

- Assets are not backed by tangible entities such as land
- Waste of computer power and technological resources.
- Enables crime
- Encourages anarchy
- It's very much still an experimental currency and is a high-risk environment for consumers and investors at the moment.

CYBER SECURITY

CRYPTOCURRENCY AND THEIR LINKAGES TO TERROR FINANCING



SUPREME COURT RULING ON CRYPTOCURRENCIES

- In a significant judgment (March 03, 2020), the Supreme Court lifted the curbs on exchanges in cryptocurrency imposed by the Reserve Bank of India.
- The Court held that the RBI's circular involving cryptocurrency was disproportionate.

CYBER SECURITY

CRYPTOCURRENCY AND THEIR LINKAGES TO TERROR FINANCING

WHY IS RBI NOT AN ADVOCATE OF CRYPTOCURRENCIES?

- Potential financial, legal and security-related risks when they deal in virtual currencies
- Losses arising out of hacking, no sources of customer recourse and the general financial volatility
- The anonymity of private digital currencies make them vulnerable to money laundering and use in terrorist financing activities while making law enforcement difficult.



CYBER SECURITY

CRYPTOCURRENCY AND THEIR LINKAGES TO TERROR FINANCING

ILLEGAL USAGE OF CRYPTOCURRENCIES IN TERRORISM



1 There are concerns about money laundering and terror financing. Besides these, all cryptocurrencies have the potential of being a **Ponzi scheme**.

2 It is also used illegally in online marketplaces that deal with **drugs and child pornography**.

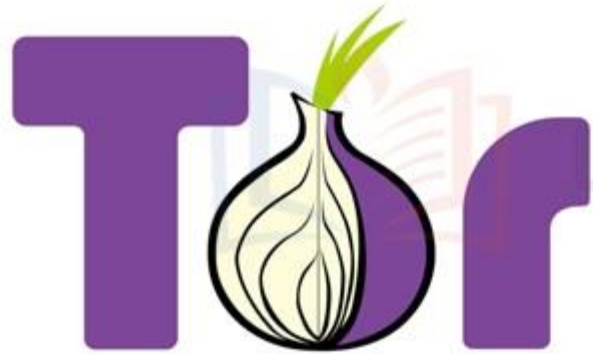
3 The money-launderers, cyber-criminals and terrorists find cryptocurrencies highly attractive because they offer **anonymity and non-traceability**.

4 It provides anonymity in the sense that the user does not have to produce any document to validate his or her identity.

CYBER SECURITY

CRYPTOCURRENCY AND THEIR LINKAGES TO TERROR FINANCING

ILLEGAL USAGE OF CRYPTOCURRENCIES IN TERRORISM



The Onion Router

5 Further, in a bitcoin network, the users are identified by an **alphanumeric value** rather than by their actual name.

6 Moreover, the transactions are **instantaneous and irreversible**. Any amount of money can be transferred instantaneously from anywhere to anywhere concerning geographical location.

7 Further, such transactions take place on the **dark web**. The dark web is an encrypted area of web with access to only specialized browsers like TOR (The Onion Router) and Freenet, and cyber criminals mostly use it, money launderers, terrorists, dissidents and journalists to avoid detection.



WORLD WIDE WEB

Only 4% of the content on the internet is www., which includes public websites such as Google, eBay, etc.

DEEP WEB

Over 90% of the information on the internet is in the deep web and is not accessible by surface web crawlers. However, it doesn't mean that they're dark web areas – they're just one layer removed from the public web that's searchable through search engines

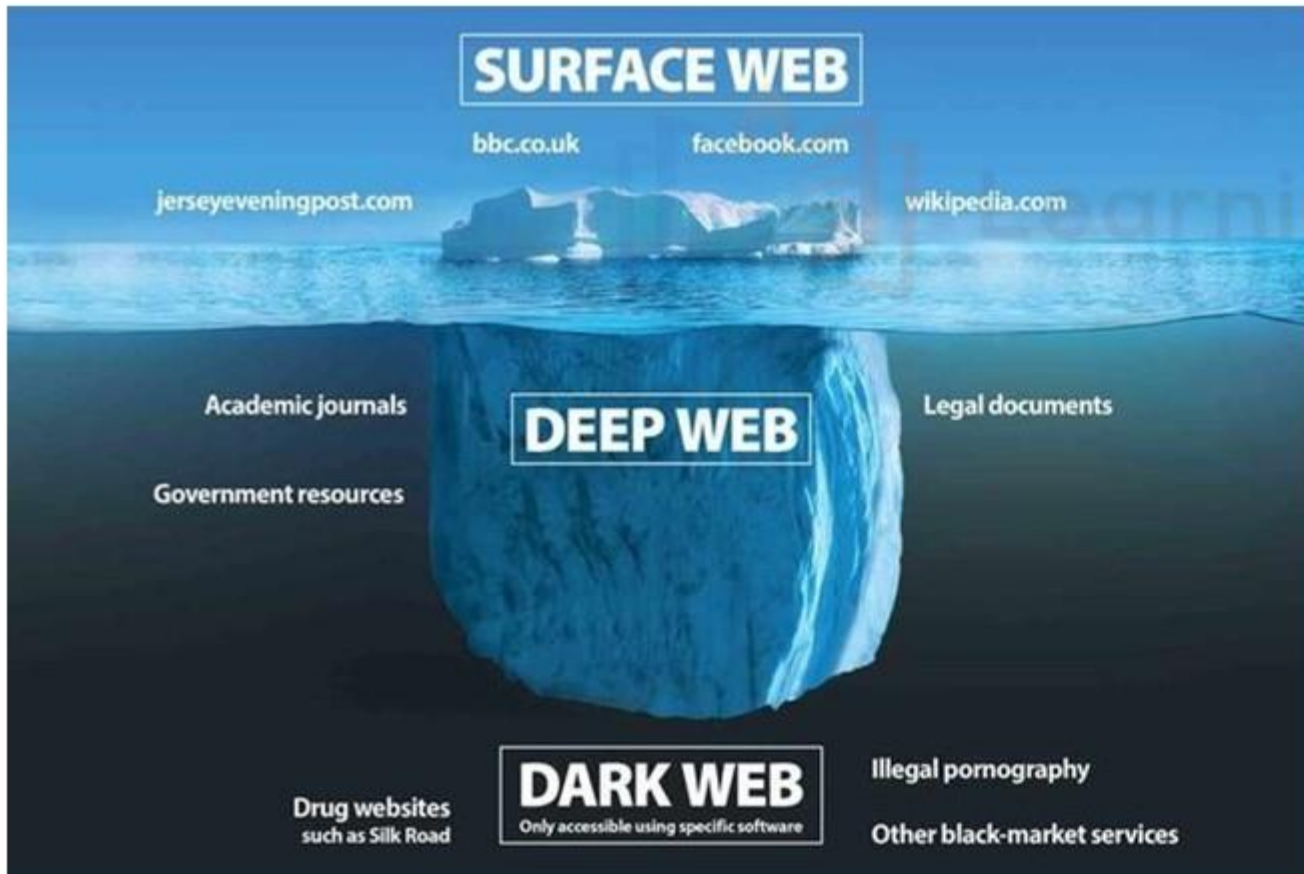
DARK WEB

The dark web consists of websites that use public internet, but require specific software for access and is not indexed by search engines to ensure anonymity. The stolen data is traded, sold and used for financial, political or personal gain.

CYBER SECURITY

CRYPTOCURRENCY AND THEIR LINKAGES TO TERROR FINANCING

ILLEGAL USAGE OF CRYPTOCURRENCIES IN TERRORISM



8 On the dark web, terrorist or cyber criminals can buy a range of things such as driving license, passport, utility bills, guns, bombs and drugs.

9 The website “Silk Road” that existed on dark web was a global network of illicit money-launderers, smuggler and all kinds of criminals.



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